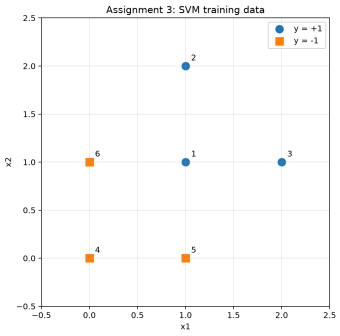
Assignment 3

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Q1: Hard-margin SVM



Support vectors: 1, 5, and 6. They are equally close to the separator; all other points are farther away.

The supporting lines have coordinate sums 1 and 2. Their midpoint sum is 1.5, giving **weights (1, 1) and bias −1.5**.

The perpendicular margin is the residual 0.5 divided by the square root of 2: **approximately 0.353553**. This is boundary-to-point distance, not the full gap.

Q2: K-means

Method. 4,000 points; four generating centers; standard deviation 0.60; seed 0 for data and initialization; 100 iterations. Points go to the nearest centroid, then centroids become cluster means. Ties favor the lowest centroid index; empty clusters retain their centroid. Initialization samples rows with replacement; true labels are unused.

Discussion. In Figure 2, two clusters merge the lower three groups; three still merge the middle pair. Four largely recover the generating groups. Six split the left-middle group into three; ten subdivide the upper and lower groups. Larger cluster counts divide existing groups rather than reveal new ones.

Q3: Noisy-label SVM and K-NN repair

Method. Seed 0; 20 paired trials on fixed, independent 2,000-point training and test sets from the specified Gaussians. Original positive/negative labels flip with probabilities 35%/20%; test labels stay clean. Both methods use RBF SVMs with C = 1 and gamma = 0.01. Repair simultaneously votes among the five nearest other points using Euclidean distance and noisy labels only; distance ties favor the lower row index.

Mean baseline accuracy	Mean after K-NN repair	Difference
82.80%	86.44%	+3.64 percentage points

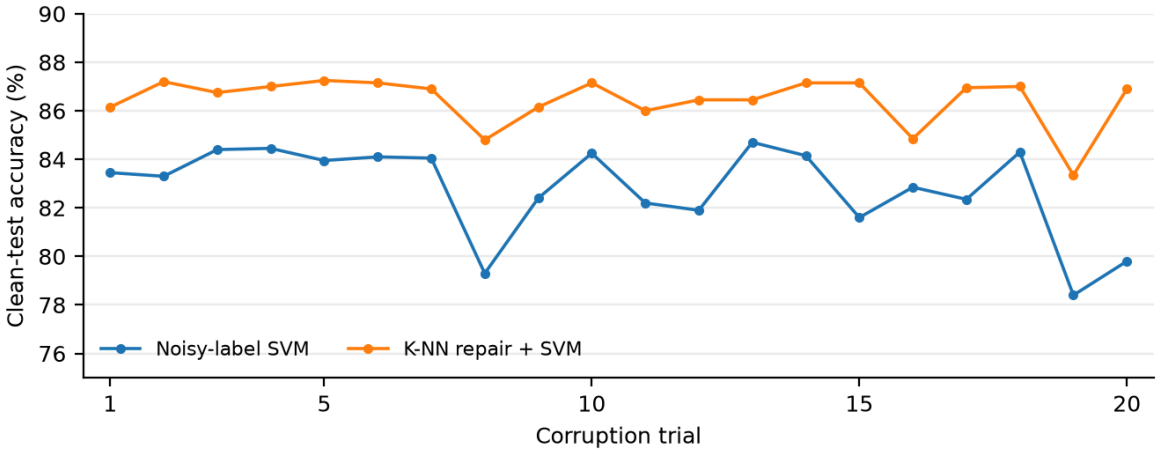


Figure 1: Clean-test accuracy across the same 20 label-corruption trials.

Discussion. Repair improved all 20 trials. Neighbor voting likely suppresses isolated label errors. However, overlapping classes or mostly corrupted neighbors can spread mistakes, so improvement is not guaranteed.

Trial data: `results/noise_comparison.csv`.

AI assistance: Q1 tutoring/code correction, experiment execution, plotting, and report drafting.

Q2: Cluster plots

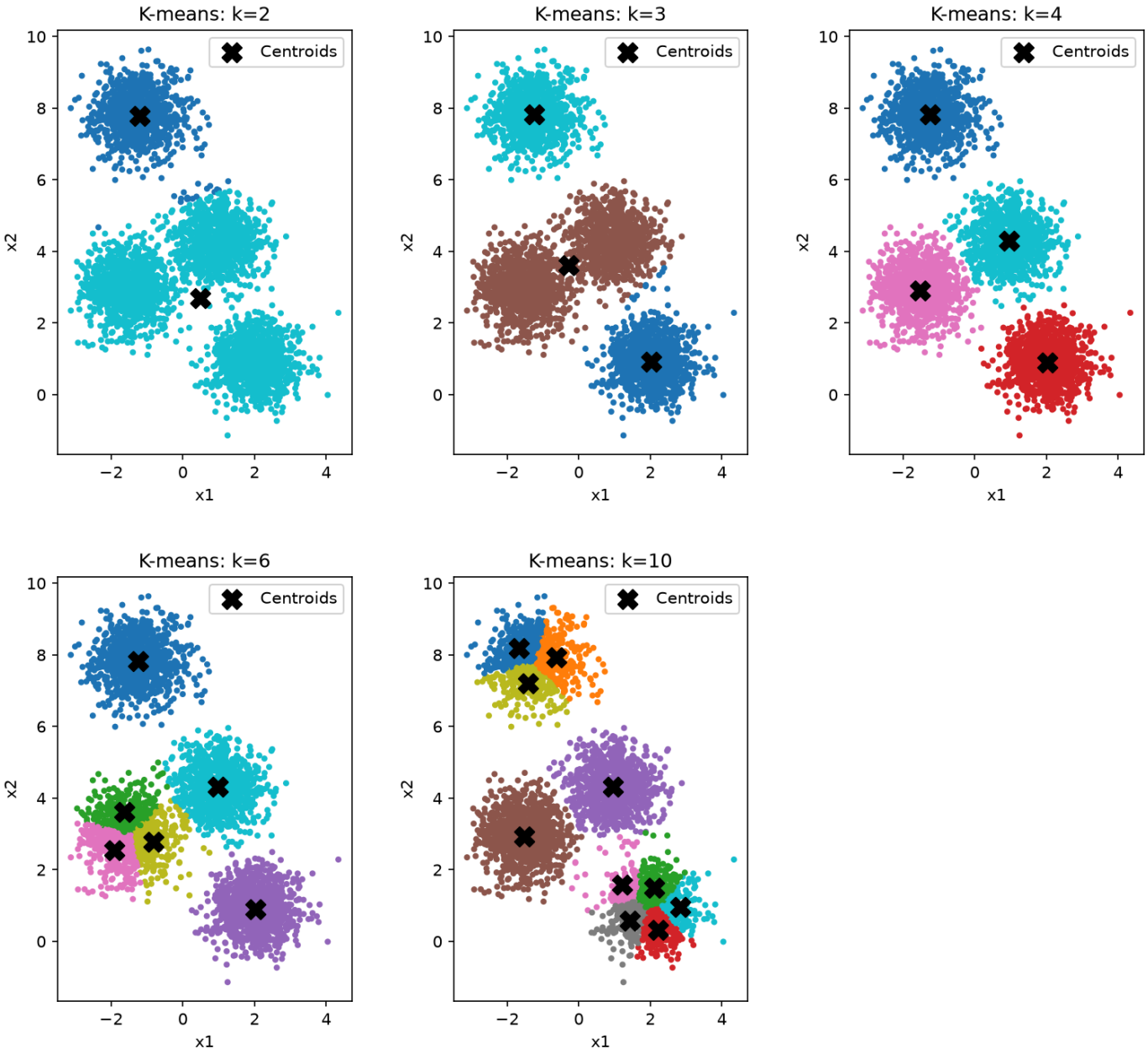


Figure 2: Assignments and final centroids for 2, 3, 4, 6, and 10 clusters. Black crosses mark centroids; colors identify clusters within each panel.